



BRAC University

Department of Mathematics and Natural Sciences

LECTURE ON

Real Analysis II (MAT321)

Topology of Metric Spaces

Open sets, Limit points and Closed sets

FEBRUARY 08, 2026

CONDUCTED BY

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Open and Closed Spheres

Open Sphere

Let (X, d) be a metric space and let $a \in X$. For any $r > 0$, the set

$$S_r(a) = \{x \in X : d(x, a) < r\}$$

is called an *open sphere* (or *open ball*) of radius r centered at a .

Closed Sphere

Let (X, d) be a metric space and let $a \in X$. For any $r > 0$, the set 

$$S_r[a] = \{x \in X : d(x, a) \leq r\}$$

is called a *closed sphere* of radius r centered at a .

Relation Between Open and Closed Spheres

Show that 

$$S_r(a) \subseteq S_r[a]$$

for every $a \in X$ and for every $r > 0$.

Neighbourhood of a Point

Neighbourhood of a Point

Let (X, d) be a metric space and let $a \in X$. A subset $N_a \subseteq X$ is called a *neighbourhood* of the point a if there exists an open sphere $S_r(a)$ centered at a such that

$$S_r(a) \subseteq N_a$$

for some $r > 0$.

② Open Sphere Is a Neighbourhood of Its Points

Show that every open sphere is a neighbourhood of each of its points.

Open Set

Open Set

A subset G of a metric space (X, d) is said to be *open* in X (with respect to the metric d) if G is a neighbourhood of each of its points. In other words, G is open if for each $a \in G$ there exists an $r > 0$ such that

$$S_r(a) \subseteq G.$$

❓ Illustrations of Open Sets

Show that each of the following subsets is open.



1. The empty set $\emptyset$ and the entire space X with any metric are open sets.
2. Every open sphere in a metric space (X, d) is an open set.
3. Let $A = \{z \in \mathbb{C} : 1 < |z| < 2\}$ with the usual metric. Then A is an open set.

4. Let

$$S = \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 < 1\}$$

with the Euclidean metric. Then S is an open set.

5. Let

$$A = \{(x, y) \in \mathbb{R}^2 : y^2 < x\}$$

with the Euclidean metric. Then A is an open set.

6. Let

$$A = \left\{ f \in C[a, b] : \inf_{x \in [a, b]} f(x) > 0 \right\}.$$

Then A is open in $C[a, b]$ with respect to the supremum metric.

② Singleton Set in $\mathbb{R}$ Is Not Open

Show that on the real line with the usual metric, the singleton set $\{x\}$ is not open.

② Singleton Sets Under Different Metrics

Let $\mathbb{R}$ be the set of real numbers, d the usual metric and d' the discrete metric on the same set $\mathbb{R}$. Show that every singleton set $\{x\}$, $x \in \mathbb{R}$, is open in $(\mathbb{R}, d')$ but not open in $(\mathbb{R}, d)$.

② Discrete Metric Space: Every Set Is Open

Show that every subset of a discrete metric space (X, d) is open. ?

② A Subset Open in a Subspace

Let (X, d) be a metric space where $X = (0, 2)$ with the usual metric. Show that the subset

$$A = (0, 1)$$

is an open set in X .

Basic Properties of Open Sets

In any metric space (X, d) ,



1. the union of an arbitrary family of open sets is open;
2. the intersection of a finite number of open sets is open.

② Intersection of Infinite Open Sets May Not Be Open

Show that the intersection of an infinite number of open sets in a metric space need not be open.

Limit Point

Limit Point

Let (X, d) be a metric space and let $A \subseteq X$. A point $x \in X$ is called a *limit point* (or *accumulation point*) of the set A if for every $r > 0$,

$$S_r(x) \cap (A \setminus \{x\}) \neq \emptyset.$$

② Illustrations of Limit Points and Derived Sets

Determine the set of limit points (derived set) in each of the following cases.

1. Let $X = \mathbb{R}$ with the usual metric and let

$$A = [0, 1).$$

Every point of A is a limit point of A , and 1 is also a limit point of A although $1 \notin A$. Hence

$$A' = [0, 1].$$

2. Let

$$A = \left\{1, \frac{1}{2}, \frac{1}{3}, \dots\right\}$$

with the usual metric in $\mathbb{R}$. Then 0 is the only limit point of A and $0 \notin A$. Hence

$$A' = \{0\}.$$

3. If (X, d) is a discrete metric space, then the derived set of every subset of X is empty, that is,

$$A' = \emptyset \quad \text{for all } A \subseteq X.$$

4. Every real number is a limit point of the set of rational numbers $\mathbb{Q}$.
Hence

$$\mathbb{Q}' = \mathbb{R}.$$

5. The set of integers $\mathbb{Z}$ has no limit point, that is,

$$\mathbb{Z}' = \emptyset.$$

6. Every finite subset of a metric space has no limit point.

7. Let

$$A = \{z \in \mathbb{C} : 1 < |z| < 2\}$$

with the usual metric. Then

$$A' = \{z \in \mathbb{C} : 1 \leq |z| \leq 2\}.$$

Closed Sets

Closed Set

A subset F of a metric space (X, d) is said to be *closed* if F contains all its limit points.

❓ Illustrations of Closed Sets

Verify the following statements concerning closed sets in metric spaces.

1. The empty set $\emptyset$ and the whole space X are closed sets in every metric space (X, d) .
2. On the real line $\mathbb{R}$ with the usual metric, the set $\mathbb{N}$ of natural numbers is closed.
3. The set $\mathbb{Q}$ of rational numbers is not closed in $\mathbb{R}$ with the usual metric.
4. Every closed interval on the real line is a closed set.
5. Every finite subset of a metric space is closed.

6. Let

$$A = \{f \in C[a, b] : f(a) = 1\}.$$

Then A is closed in $C[a, b]$ with respect to the supremum metric.

7. The set

$$\left\{ \frac{1+i}{n} : n \in \mathbb{N} \right\}$$

is neither open nor closed in the complex plane with respect to the usual metric.

Equivalent Definition of Closed Sets

Let (X, d) be a metric space. A subset $F \subseteq X$ is closed if and only if its complement $X \setminus F$ is open. 

② Closed Sphere Is a Closed Set

Show that every closed sphere in a metric space (X, d) is a closed set.

Basic Properties of Closed Sets

In any metric space (X, d) ,



1. the intersection of an arbitrary family of closed sets is closed;
2. the union of a finite number of closed sets is closed.

② Union of Infinite Closed Sets May Not Be Closed

Show that the union of an infinite number of closed sets in a metric space need not be closed.

Closure of a Set

Closure

Let A be a subset of a metric space (X, d) . The *closure* of A , denoted by $\overline{A}$ is the smallest closed set containing A , that is,

$$\overline{A} = \bigcap \{F : F \text{ is closed and } A \subseteq F\};$$

Equivalent Definition of Closure

Let A be a subset of a metric space (X, d) . Then

$$\overline{A} = A \cup A'.$$



❓ Illustrations of Closure

Determine the closure of each of the following subsets.

1. Let $X = \mathbb{R}$ with the usual metric and let

$$A = (0, 1) \cup (1, 2).$$

Then

$$\overline{A} = [0, 2].$$

2. Let $X = \mathbb{R}$ with the usual metric and let

$$A = \left\{ \frac{1}{n} : n \in \mathbb{N} \right\}.$$

Then

$$\overline{A} = A \cup \{0\}.$$

3. If (X, d) is a discrete metric space and $A \subseteq X$, then

$$\overline{A} = A.$$

4. Let $X = \mathbb{N}$ with the usual metric and let $A = \{1, 2, 3\}$. Then

$$\overline{A} = A.$$

💡 Properties of Closure

Let (X, d) be a metric space and let A, B be any two subsets of X .
Then:

1. A is closed if and only if $A = \overline{A}$;
2. $A \subseteq B$ implies $\overline{A} \subseteq \overline{B}$;
3. $\overline{A \cup B} = \overline{A} \cup \overline{B}$;
4. $\overline{A \cap B} \subseteq \overline{A} \cap \overline{B}$.
5. $\overline{\overline{A}} = \overline{A}$.
6. $\overline{A} = (\text{ext } A)^c$.
7. $\overline{A} = \text{int}(A^c)^c$.

Thank You!

We'd love your questions and feedback.

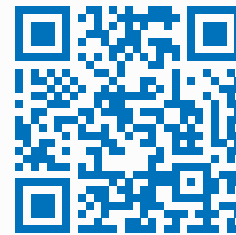
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(Lectures, walkthroughs, and course updates)



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References

- [1] Erwin Kreyszig, *Introductory Functional Analysis with Applications*, John Wiley & Sons, 1978.
- [2] Walter Rudin, *Principles of Mathematical Analysis*, 3rd Edition, McGraw-Hill, 1976.